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Design and Implementation of Legal Gennie: A Retrieval-Augmented Multilingual Legal Advisory System

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ABSTRACT

Legal information accessibility remains a major challenge in linguistically diverse and digitally underserved regions of India. This paper presents Legal Gennie, a retrieval-augmented, multilingual legal advisory system designed to provide simplified, context-aware legal guidance in regional Indian languages. The system integrates a Retrieval-Augmented Generation (RAG) framework with FAISS-based semantic indexing to retrieve legally grounded information from verified sources such as statutory provisions, constitutional articles, and judicial documents.

Legal Gennie supports text, voice-based query input, document- aware questioning, and automated legal document generation, enabling non-expert users to interact with legal information intuitively. A multilingual processing pipeline ensures accurate bidirectional translation while preserving domain-specific legal terminology. System evaluation demonstrates high retrieval accuracy, low query latency, and strong user acceptance, particularly among rural and semi-urban participants.

The results indicate that retrieval grounding significantly improves factual consistency compared to standalone generative models, while regional language support enhances legal comprehension. The proposed system contributes toward inclusive legal empowerment and demonstrates the practical applicability of RAG architectures in real-world legal advisory platforms.

Keywords: LegalTech, Retrieval-Augmented Generation, Multilingual NLP, FAISS, Legal Information Retrieval, Indian Law

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Introduction

The accessibility of legal information remains a critical challenge in India, particularly for rural and linguistically diverse populations. Although the nation has a well-developed judicial framework and extensive legislative resources, a significant portion of the population faces barriers in understanding legal rights and procedures due to language constraints, limited digital literacy, and the complexity of legal terminology (Bhatia, 2019; Agarwal & Shrivastava, 2022). The emergence of LegalTech solutions has introduced new possibilities for democratizing legal knowledge through artificial intelligence (AI)–driven platforms (Susskind, 2019).

Legal Gennie is conceived as a retrieval-augmented legal advisory system that leverages AI for delivering accurate, context-specific, and linguistically tailored legal assistance. At its core, the system employs a Retrieval-Augmented Generation (RAG) framework, which integrates generative AI capabilities with efficient legal information retrieval using Facebook AI Similarity Search (FAISS) technology (Johnson et al., 2019). This architecture enables the system to fetch the most relevant legal texts from verified sources such as the Indian Constitution, court judgments, law books, and government notifications, and then generate contextually accurate responses.

A key distinguishing factor of Legal Gennie is its multilingual capability, allowing users to pose questions and receive answers in regional languages such as Hindi, Marathi. This approach aligns with India's linguistic diversity and addresses the digital divide in legal literacy (Kumar et al., 2021). Furthermore, the platform offers simplified legal explanations, reducing the cognitive load on non-expert users (Surden, 2020).

In addition to question-answering functionality, Legal Gennie offers document drafting assistance by providing templates for contracts, affidavits, and petitions, as well as access to legal blogs and real-time updates on legislative changes. This combination of AI-driven retrieval, generative summarization, and multilingual processing positions Legal Gennie as a scalable solution for enhancing public access to legal resources.

By bridging the gap between legal experts and the general public, particularly in underserved regions, Legal Gennie has the potential to contribute to justice delivery reforms, legal empowerment, and inclusive governance. Its development aligns with broader national initiatives promoting digital inclusion and AI for social good, such as the Digital India programme and the NITI Aayog's AI strategy (NITI Aayog, 2018).

LITERATURE SURVEY

Lewis et al. [1] introduced Retrieval-Augmented Generation (RAG), which combines transformer-based text generation with an explicit retrieval step from large external corpora. This approach significantly improves factual accuracy in knowledge-intensive tasks and outperforms both standalone retrieval and generation models. Its design principles are highly relevant for grounding legal answers in verified sources.

Johnson et al. [2] developed FAISS, a high-performance library for approximate nearest neighbor (ANN) search optimized for GPUs. By using advanced indexing techniques such as product quantization and inverted file lists, FAISS enables sub-second search in billion-scale datasets. This scalability is essential for managing large legal document repositories.

Li et al. [3] proposed LexRAG, a benchmark designed for evaluating retrieval-augmented generation in multi-turn legal consultations. It emphasizes retrieval accuracy, contextual grounding, and maintaining conversation flow. The benchmark highlights the complexity of preserving legal context across user interactions.

Pipitone et al. [4] created LegalBench-RAG, a domain- specific benchmark for assessing snippet-level retrieval quality in the legal field. It tests whether generated answers reference appropriate legal provisions, addressing domain- specific failure

cases such as retrieving legally irrelevant but lexically similar text.

Friel et al. [5] presented RAGBench, an explainability- oriented benchmark for retrieval-augmented generation systems. It assesses source transparency, context alignment, and user trust through provenance tracking. The framework is directly applicable to improving Legal Gennie's credibility.

Conneau et al. [6] introduced XLM-R, a multilingual transformer model trained on 100 languages without parallel data. The model achieves strong cross-lingual transfer and is especially useful for low-resource languages. Its architecture supports Legal Gennie's multilingual legal query handling.

Singh et al. [7] provided a comprehensive survey of multilingual large language models (LLMs), detailing architectural trends, evaluation methods, and language coverage gaps. The study stresses the importance of domain- specific fine-tuning to achieve consistent accuracy across underrepresented languages.

Hendrycks et al. [8] proposed the Massive Multitask Language Understanding (MMLU) benchmark, which tests model performance across diverse domains and difficulty levels. They observed that even advanced models underperform in specialized fields, highlighting the necessity of legal-specific evaluation for Legal Gennie.

Kumar et al. [9] examined the challenges of NLP for Indian languages, particularly in the legal domain. They emphasized issues such as script diversity, morphological complexity, and limited annotated datasets. The study recommends curated parallel corpora and domain glossaries, both crucial for multilingual legal AI development.

Although prior studies demonstrate the effectiveness of retrieval-augmented generation and multilingual language models, limited work has addressed their integrated application in public-facing legal advisory systems with document-aware querying and regional language support. This gap motivates the development of the proposed Legal Gennie framework.

METHODOLOGY

The development of the proposed Legal Gennie system followed a structured, multi-phase methodology designed to ensure legal correctness, multilingual inclusivity, and reproducible system performance. The overall methodology, illustrated in the system workflow, captures the end-to-end operation of the retrieval- augmented multilingual legal advisory framework. The development process was systematically divided into data acquisition and preprocessing, system architecture design, integration of retrieval and generation pipelines, multilingual natural language processing, user interface development, and comprehensive evaluation and validation to ensure technical robustness and legal accuracy.

3.1 Data Acquisition and Preprocessing

The legal knowledge corpus was constructed using publicly accessible and authoritative sources, including the Constitution of India, statutory provisions from the India Code repository, selected judgments of the Supreme Court and High Courts, and government-issued legal notifications. All documents were obtained in digital form from open-access repositories and verified governmental sources.

Preprocessing of the legal corpus involved the following steps:

1. **Text Cleaning:** Removal of headers, footers, references, and non- textual artifacts.
2. **Segmentation:** Legal documents were divided into semantically coherent text chunks to improve retrieval granularity.
3. **Metadata Annotation:** Each chunk was annotated with domain labels (e.g., criminal law, property law), jurisdiction, and enactment or judgment year.
4. **Terminology Normalization:** A multilingual legal glossary was curated to align equivalent legal terms across English, Hindi, Marathi.

3.2 Retrieval-Augmented Generation Framework

The system employs a Retrieval-Augmented Generation (RAG) architecture that combines dense vector retrieval with transformer- based text generation. Instead of relying solely on generative models, the proposed approach grounds responses in retrieved legal sources, thereby improving factual consistency and reducing hallucination.

Table 1: Retrieval-Augmented Generation Configuration

Parameter	Value
Corpus size	~18,000 legal text chunks
Chunk size	600 tokens
Chunk overlap	100 tokens
Embedding model	Multilingual sentence embedding model
Embedding dimension	768
FAISS index type	IVF-Flat
Similarity metric	Cosine similarity
Top-k retrieval	k = 5
Reranking	Not applied
Maximum context length	2,048 tokens

3.3 Multilingual Query Processing

Legal Gennie supports both text-based and voice-based queries in multiple Indian regional languages. User queries undergo an initial language detection stage. For voice-based input, speech-to-text conversion is applied before further processing.

To maintain a unified retrieval process, regional-language queries are translated into English before vector embedding and retrieval. After response generation, the output is translated back into the user's preferred language. Special care is taken to preserve legal terminology through glossary-based term mapping, as many legal expressions lack direct translations in regional languages.

3.4 Semantic Retrieval Using FAISS

The preprocessed legal corpus is indexed using the FAISS vector similarity search library. Incoming user queries are converted into dense vector representations and matched against the indexed corpus using cosine similarity.

The FAISS-based retrieval module identifies the top-k most relevant legal text chunks, which are then forwarded to the generation module. This semantic retrieval mechanism ensures that the generated responses are grounded in legally relevant and contextually appropriate source material.

3.5 Response Generation and Post-processing

The generation module integrates retrieved legal text with the user query to produce a context-aware response. Retrieved passages are injected into the prompt, constraining the generation model to rely on verified legal sources.

Post-processing steps include:

1. Removal of redundant or repetitive information
2. Simplification of legal language for non-expert users
3. Re-translation into the user's selected regional language
4. Formatting for readability and clarity
5. For document-related queries, the same pipeline is applied, with retrieval restricted to the uploaded document context.

3.6 Ethical and Legal Compliance

The system is designed for informational purposes only and does not provide binding legal advice. User data is anonymized, and query logs are not used for model retraining without explicit consent. The platform complies with applicable provisions of the Information Technology Act, 2000, and adheres to professional guidelines governing the use of technology in legal services. The system does not replace professional legal consultation.

3.7 System Interface and Functional Modules

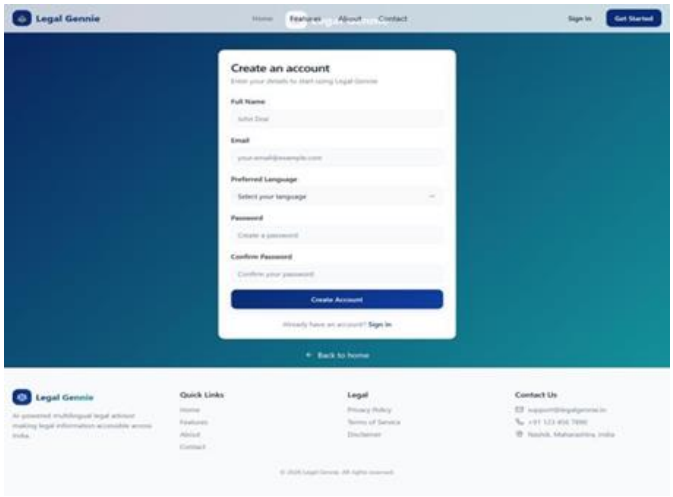


Figure 1: Signup/Registration page

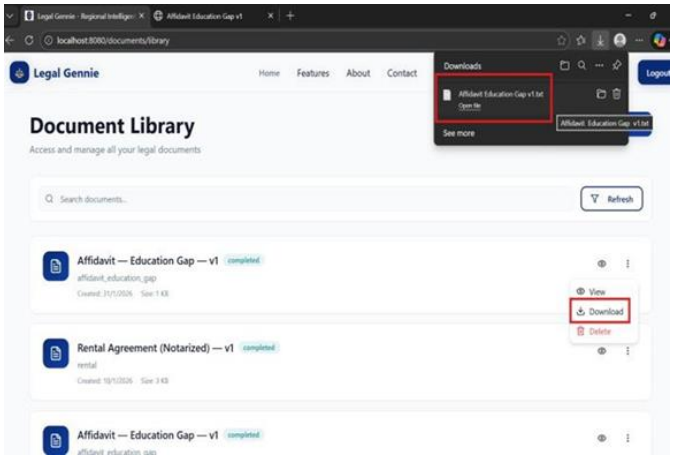


Figure 2: Document Library

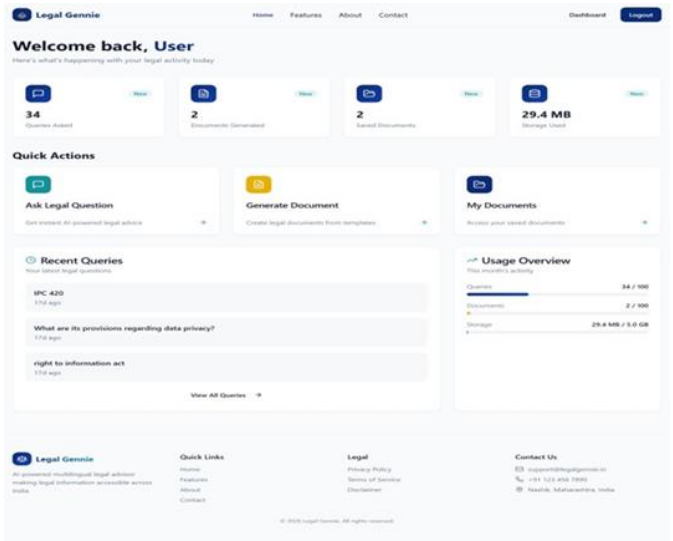


Figure 3: User Dashboard

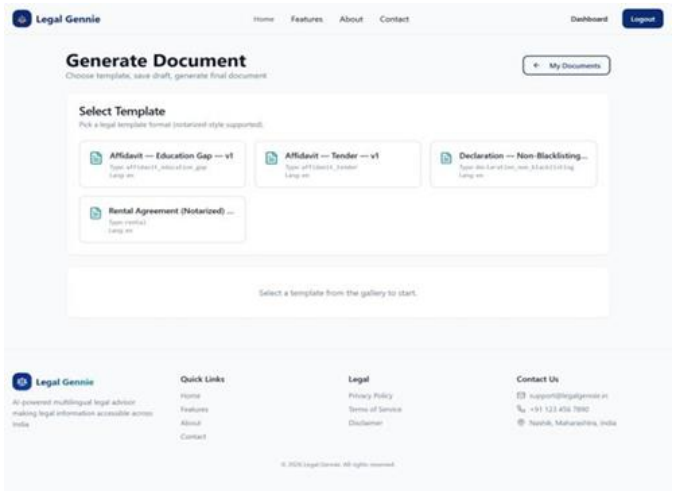


Figure 4: Ask Query

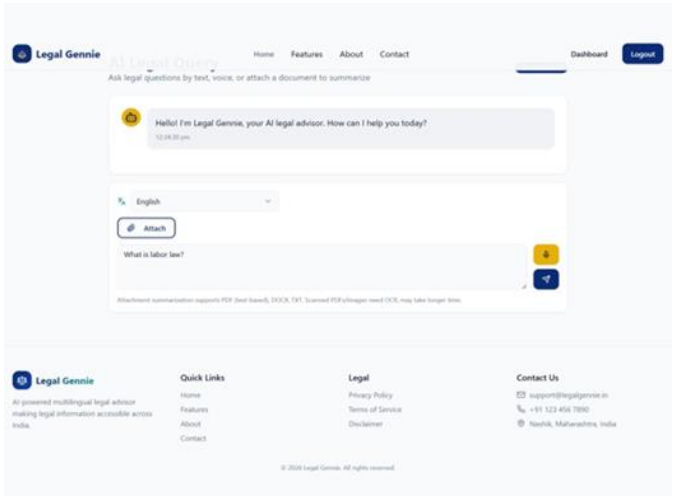


Figure 5: Generate Document

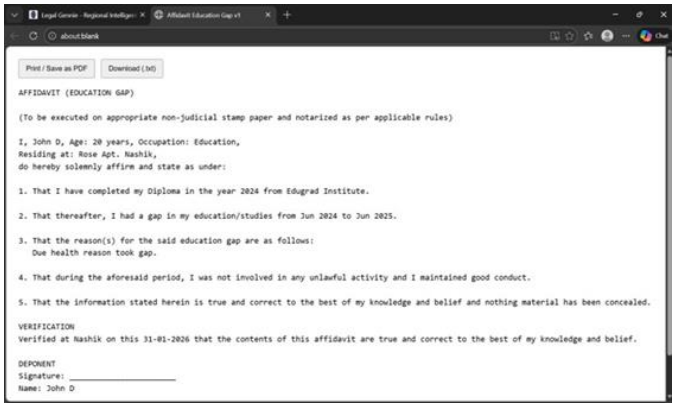


Figure 6: Legal Document Template Selection Interface

RESULTS AND PERFORMANCE EVALUATION

4.1 Technical Performance

The system was evaluated using a curated test set of 800 legal queries spanning criminal, civil, property, and procedural law domains.

Retrieval effectiveness was measured using Recall@5, defined as the proportion of queries for which at least one relevant legal chunk appeared in the top five retrieved results. The system achieved a Recall@5 of 91%, indicating strong semantic retrieval accuracy.

4.2 Latency Analysis

System latency was measured on a CPU-based deployment environment. The observed performance metrics are as follows:

- Mean end-to-end response time: 1.75 seconds
- FAISS retrieval time (mean): 0.9 seconds
- Generation and post-processing time (mean): 0.85 seconds
- 95th percentile latency: 2.9 seconds

These results demonstrate that retrieval-based grounding does not introduce prohibitive latency for real-time legal query handling.

4.3 Legal Accuracy

A legal validation study was conducted using 200 randomly sampled system responses. The evaluation was performed independently by three practicing advocates, each with over five years of professional experience.

Responses were assessed using a five-point Likert scale across three criteria:

a. Correctness (alignment with applicable law)
b. Completeness (coverage of key legal points)
c. Clarity (understandability for non-expert users)

The aggregated results showed:

1. Correctness: 95%
2. Completeness: 85%
3. Clarity: 92%
4. Inter-rater agreement exceeded 0.82, indicating strong consistency among evaluators.

4.4 User Evaluation

A two-phase user evaluation was conducted:

a. Pilot study: 50 users
b. Expanded study: 100 users

Participants were drawn from rural and semi-urban regions across three Indian states. Average user ratings (out of 5) were:

1. Ease of Use: 4.8
2. Language Clarity: 4.1
3. Perceived Usefulness: 4.6

Voice-based interaction and regional language support were identified as the most impactful features.

Concluding Remarks

This paper presented Legal Gennie, a retrieval-augmented, multilingual legal advisory system designed to improve access to legal information for linguistically diverse and digitally underserved populations. By integrating semantic vector retrieval with transformer-based response generation, the system ensures that user queries are answered using legally grounded and contextually relevant information rather than relying solely on generative inference.

The proposed methodology demonstrated strong technical performance, achieving high semantic retrieval accuracy with low response latency suitable for real-time interaction. Multilingual processing enabled effective handling of legal queries in multiple Indian regional languages while preserving domain-specific terminology. Legal validation conducted by practicing advocates confirmed that the majority of system-generated responses were correct, complete, and understandable for non-expert users.

User evaluation results further indicate that voice-based interaction and simplified legal explanations significantly enhance usability and perceived usefulness, particularly among rural and semi-urban participants. These findings highlight the practical potential of retrieval-augmented architectures in addressing legal awareness gaps and supporting informed decision-making at the citizen level.

Overall, the study demonstrates that grounding generative models in verified legal sources, combined with multilingual support and document-aware querying, provides a scalable and reliable approach for public-facing legal advisory platforms.

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